Sequence diagram — AI assistance (1/2): maintenance ticket triage

Front end · React

Reporter

MaintenancePage

Back end · NestJS

AiController

«service» **MaintenanceTriage Service**

«service» **OpenAiService**

«repository» **PrismaService**

Data

PostgreSQL

External

«external» **LLM provider (OpenAI-compatible)**

1 title and description

2 POST /ai/maintenance/triage

3 triage(user, { title, description })

1. category and priority

opt [API key configured]

4 chatJson(text, allowed categories and priorities)

5 POST /chat/completions

6 { category, priority, reason }

7 parsed answer, or null

alt [both values belong to the enumerations]

8 source = "model"

[invented value, no answer, or no key]

9 keyword classifier, source = "keywords"

2. suggested technician and past fixes

opt [caller can assign (SUPER_ADMIN, CLIENT_ADMIN, MANAGER)]

10 active technicians, and their resolved and open tickets in this category

11 SELECT users, maintenance_tickets

12 score = 10 × resolved − 6 × open (tie: lighter load first)

13 3 latest resolved tickets of this category in the company's buildings

3. the reporter decides

14 category, priority, reason, suggestedAssignee, similar, source

15 201 Created

16 show the suggestion

opt [the reporter accepts it]

17 Apply / Assign to them

18 fill in category, priority, assignee

ref

Submit the ticket: POST /maintenance → MaintenanceService.create():
403 for a renter whose company neither booked nor leased the space;
otherwise ticket ASSIGNED (assignee given) or OPEN, space MAINTENANCE
if URGENT or EMERGENCY, assignee notified and e-mailed.

The model proposes; it never decides. The technician comes from ticket history, and submitting a ticket never depends on the model.